Open-source Pulseq Sequences with United Imaging

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Background

- Develop sequence
- Publish paper (high efficiency)
- Low entry barrier

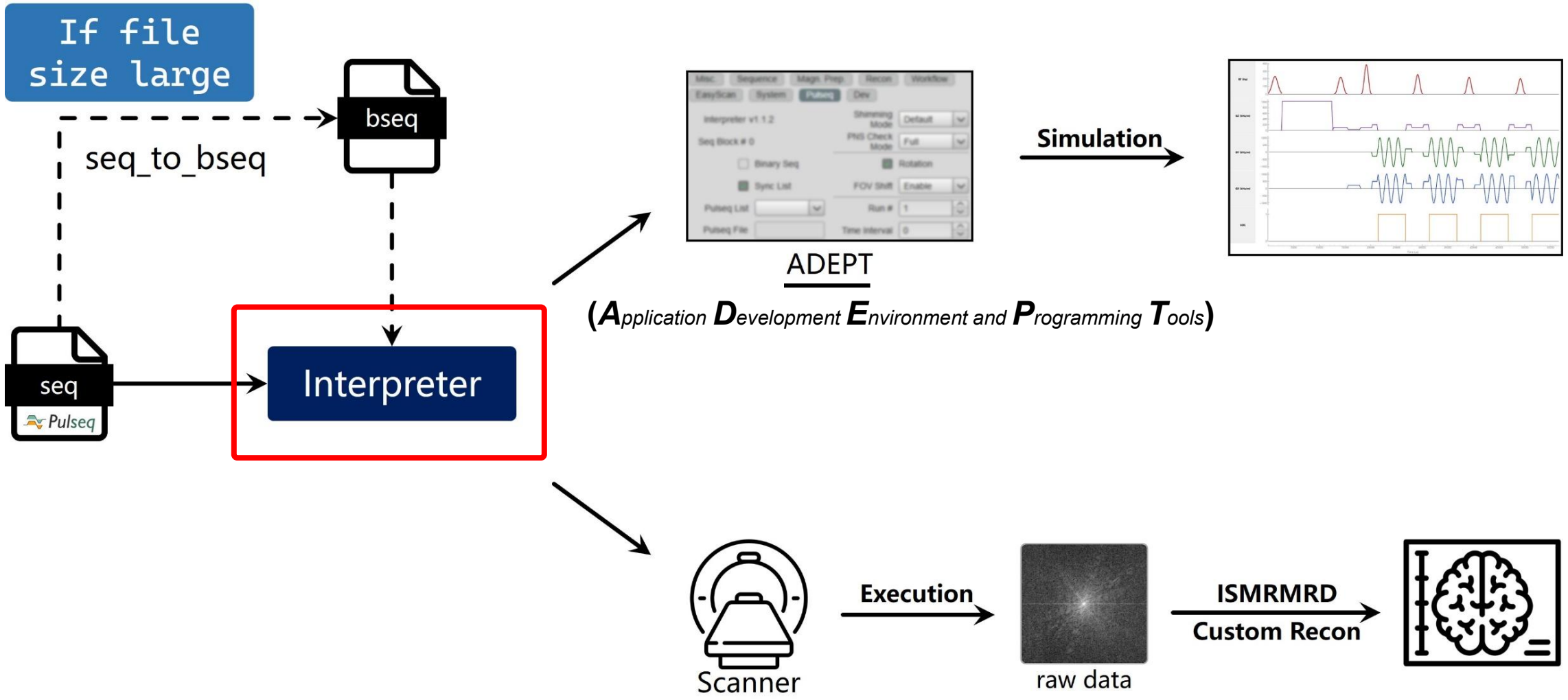
- Engineering design (High entry barrier)
- Differences among vendors' software (Multiple coding)
- Differences among vendors' seq implementation

Researchers

MR Vendor

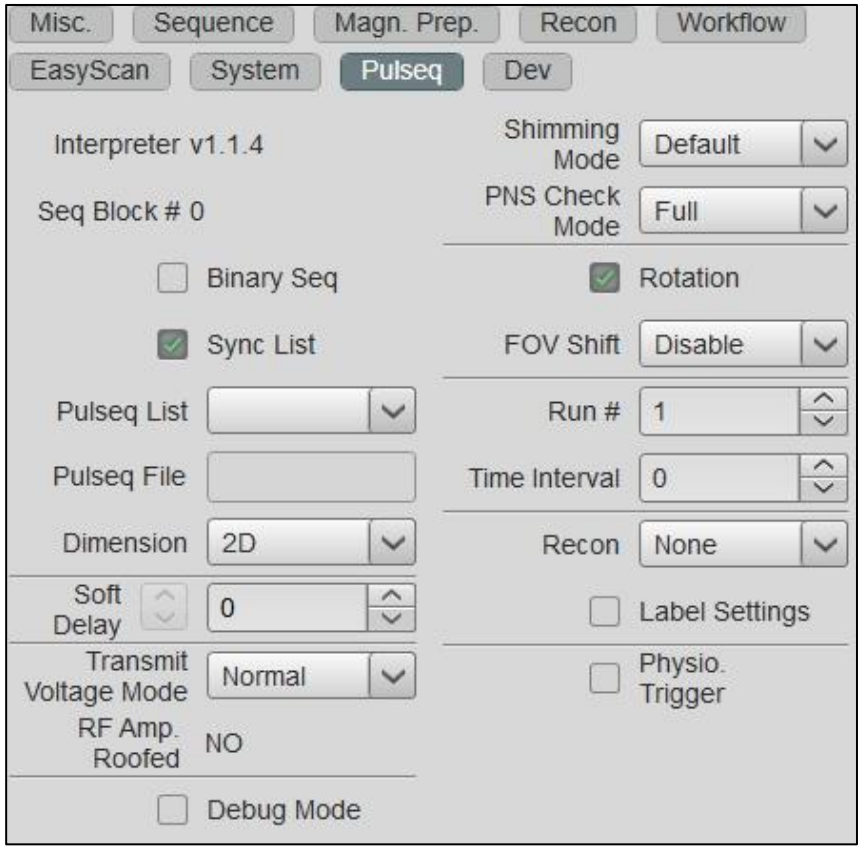
Vendors' MR sequences development solutions are not well-suited for scientific researchers

Sequence Development Workflow



Interpreter Features

- Fully support of Pulseq v1.5.0 (v1.5.1 WIP)
- **Arbitrary** sequence waveform design (GRE, SE, EPI, MRS)
- Physio. trigger and soft delay
- **FOV positioning**
- Repeated run
- Nice debug settings: scaling RF or gradients
- Friendly **binary seq format** for large sequence files (e.g., ultra-high resolution)
- **Gadgetron** compatible —— free workflow for researchers



1.5T ~ 9.4T Available on all products

Binary seq format (.bseq)

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Binary seq format (**.bseq**) is designed to solve
1. excessively long loading time
2. potential execution failure

All sequence files used for testing had a .seq file size exceeding **5MB** and contained more than **50,000 blocks**.

Sequence	.seq file size (MB)	Block num	Arb grad	.seq load time (s)	.bseq load time (s)	Time reduction (%)	.seq mem usage (GB)	.bseq mem usage (GB)	Mem reduction (%)
4D flow	458	512768	Yes	50.2	0.98	98.05	2.20	0.61	72.27
GRE 3D	55.9	1469441	No	15.6	0.5	96.79	2.15	0.56	73.95
Radial 3D	42.0	1217966	No	11.4	0.38	96.67	1.69	0.43	74.56
MIMOSA[13, 14]	16.7	487869	No	4.6	0.14	96.96	0.61	0.16	73.77
MWF qalas[15, 16]	8.2	245950	No	2.3	0.08	96.52	0.33	0.07	78.79
PRIME[17-19]	7.7	77870	Yes	1.9	0.03	98.42	0.2	0.03	85

Loading time: over **95%** reduction

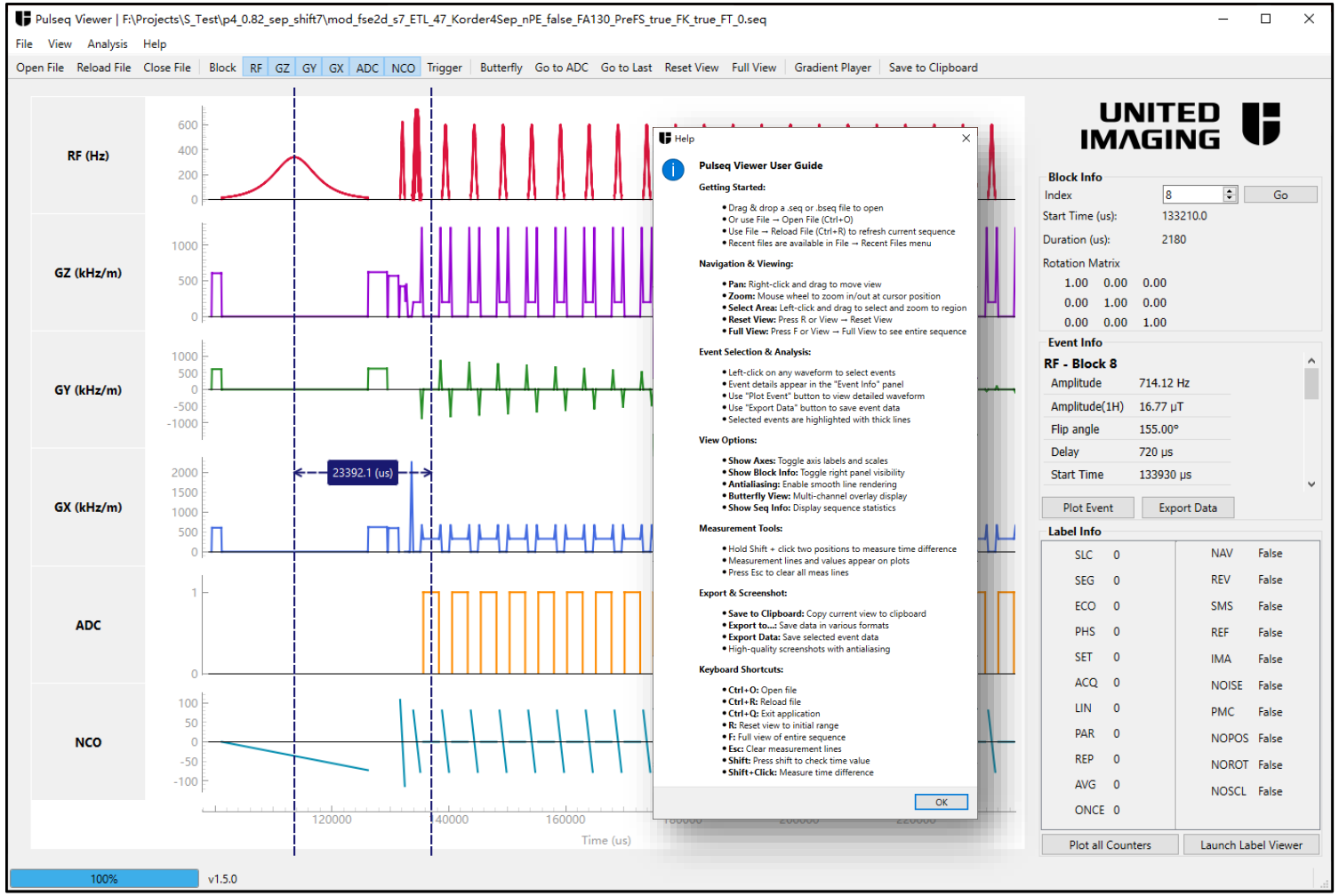
Memory usage: over **70%** reduction

All tests based on Pulseq 1.5.0

2026 Abstract **#00657**: Open-source Pulseq Sequences with United Imaging MRI Systems

Pulseq Viewer

- **Event property calculation**
- **Time distance measurement**
- **View sequence labels**
- **UI modification**
  - Zooming
  - Panning
- **Sequence compare**
- **Sequence metrics**
  - min dwell
  - max grad amp & slew
  - some other max/min properties
- **Gradient player**
- **And more …**



Online Recon with Gadgetron

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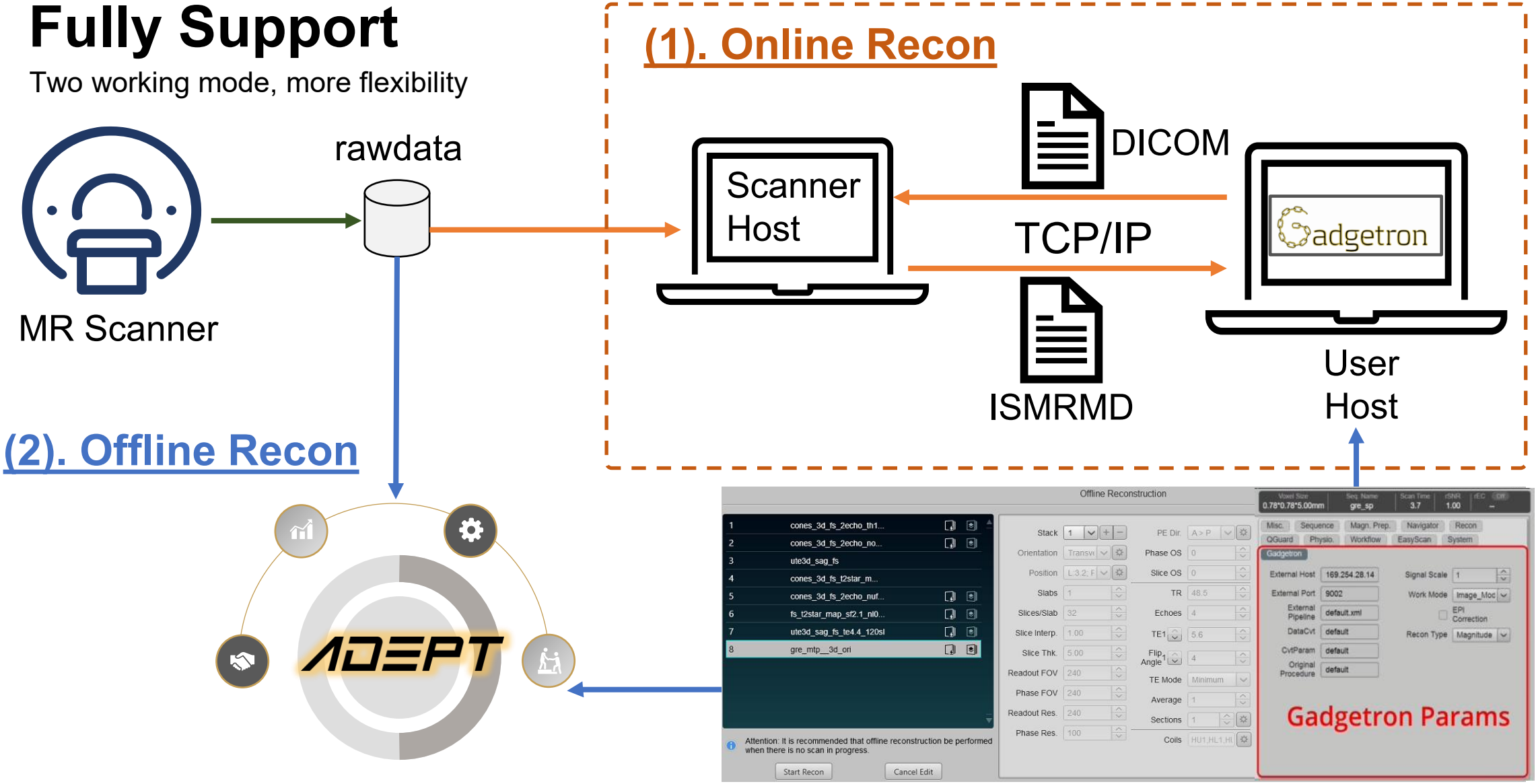
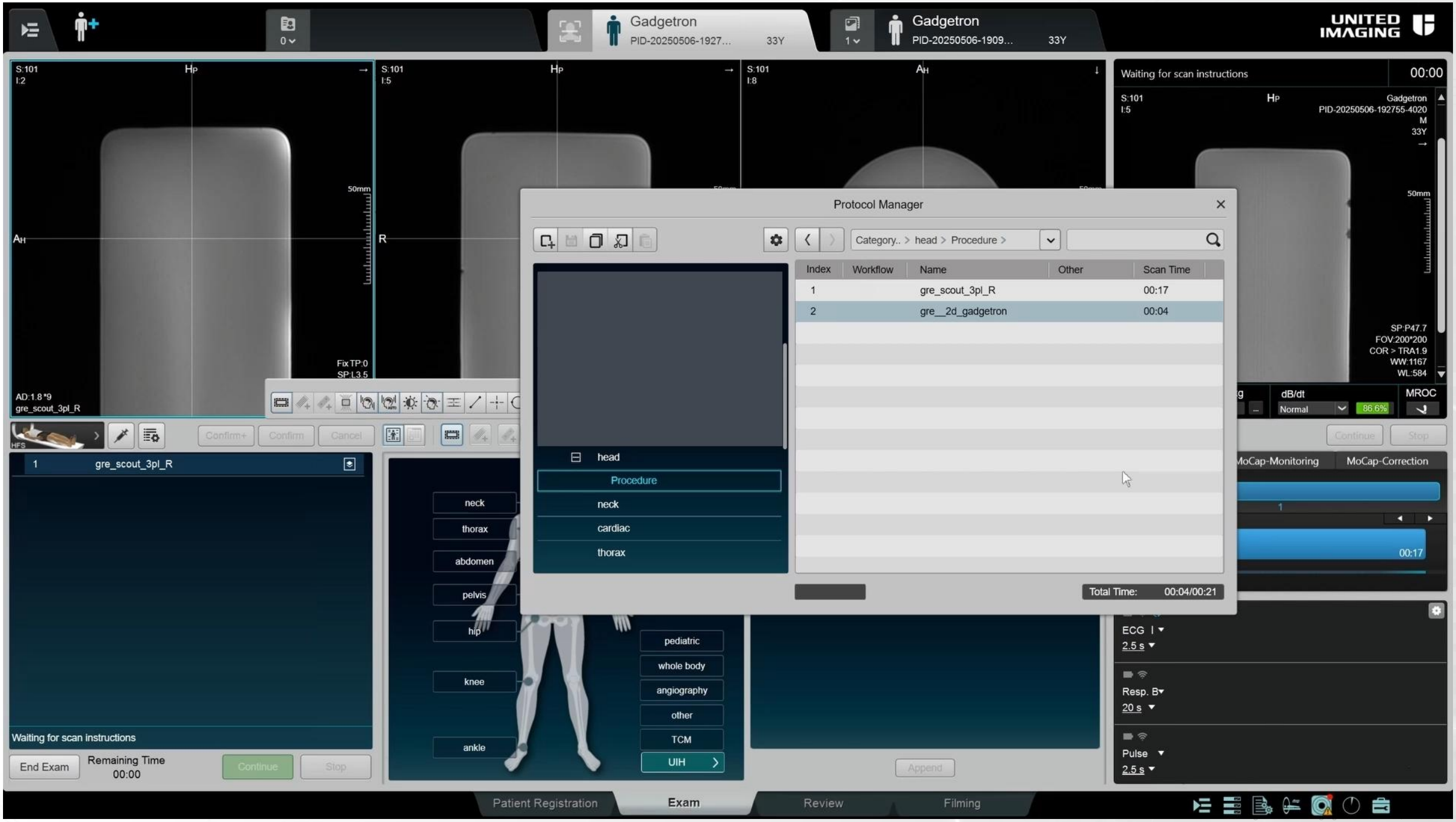
Fully Support

Two working mode, more flexibility




Gadgetron
PID-20250506-1927...
33Y
Gadgetron
PID-20250506-1909...
33Y
UNITED IMAGING
Waiting for scan instructions
00:00
Protocol Manager
Category.. > head > Procedure >
Index
Workflow
Name
Other
Scan Time
1
gre_scout_3pl_R
00:17
2
gre__2d_gadgetron
00:04
head
Procedure
neck
cardiac
thorax
Total Time: 00:04/00:21
neck
thorax
abdomen
pelvis
hip
knee
ankle
pediatric
whole body
angiography
other
TCM
UIH
gre_scout_3pl_R
Waiting for scan instructions
End Exam
Remaining Time
00:00
Continue
Stop
Confirm+
Confirm
Cancel
Append
MoCap-Monitoring
MoCap-Correction
ECG
Resp. B
Pulse
Patient Registration
Exam
Review
Filming

Key details to consider

- Careful with RF transmission voltage
- RF samples < 4096
- Arbitrary gradient samples < 30000
- ADC samples < 10240
- ADC dwell time: integer multiple of 100ns

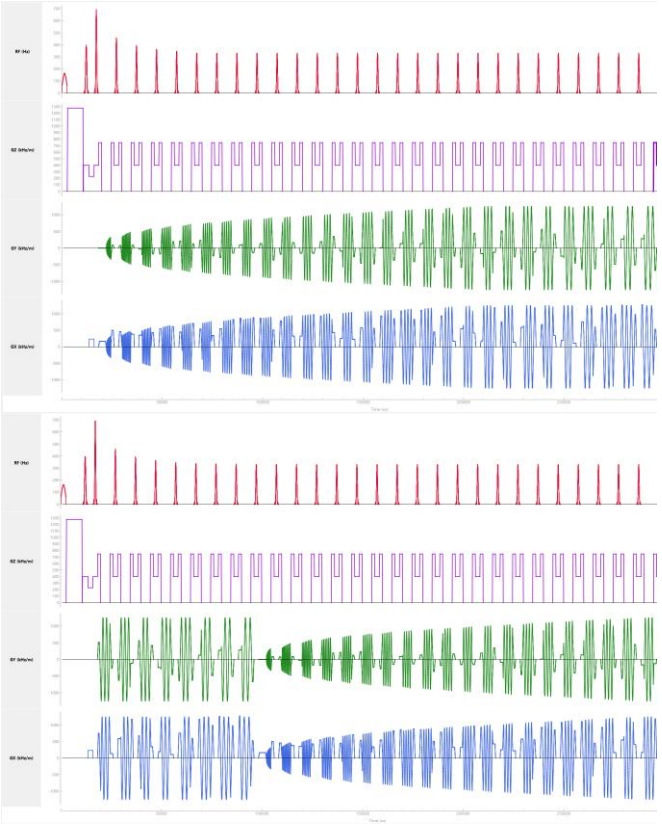
Arbitrary gradient spec diff

UIH	Pulseq	Interpreter
Arb grads in the same STB should be in one RTO if overlapping in time	Fully independent	Set all axis arb grads in one RTO
Uniform sampling	Allow non-uniform	resampling
Start and end at 0	unnecessary	Add points to ramp down to zero if necessary
All arb grads in the same STB are time-aligned if overlapping in time	unnecessary	Add zero points manually

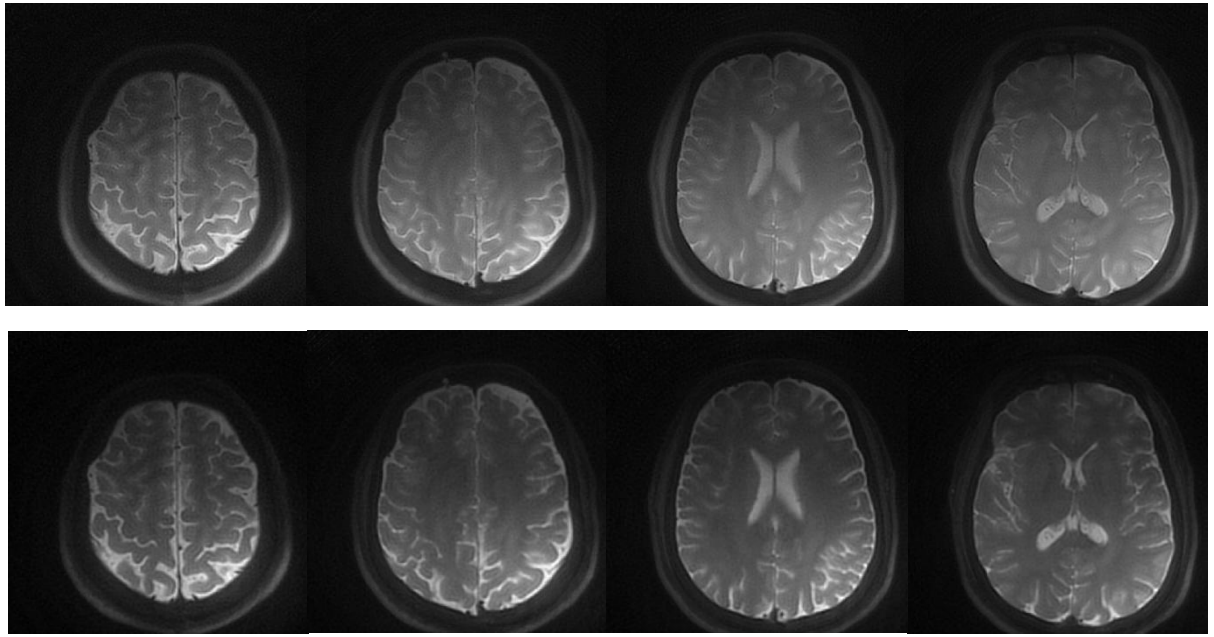
Showcase 1

Single-shot spiral TSE pulseq sequence at uMR 3T[1].

[Resolution=1×1×4mm, FOV=240×240, ESP = 10ms, ETL = 28, TE_eff=10/90ms, TR=1000ms]



Single-shot spiral TSE sequence waveform.
Echo permutation is used to control effective TE.



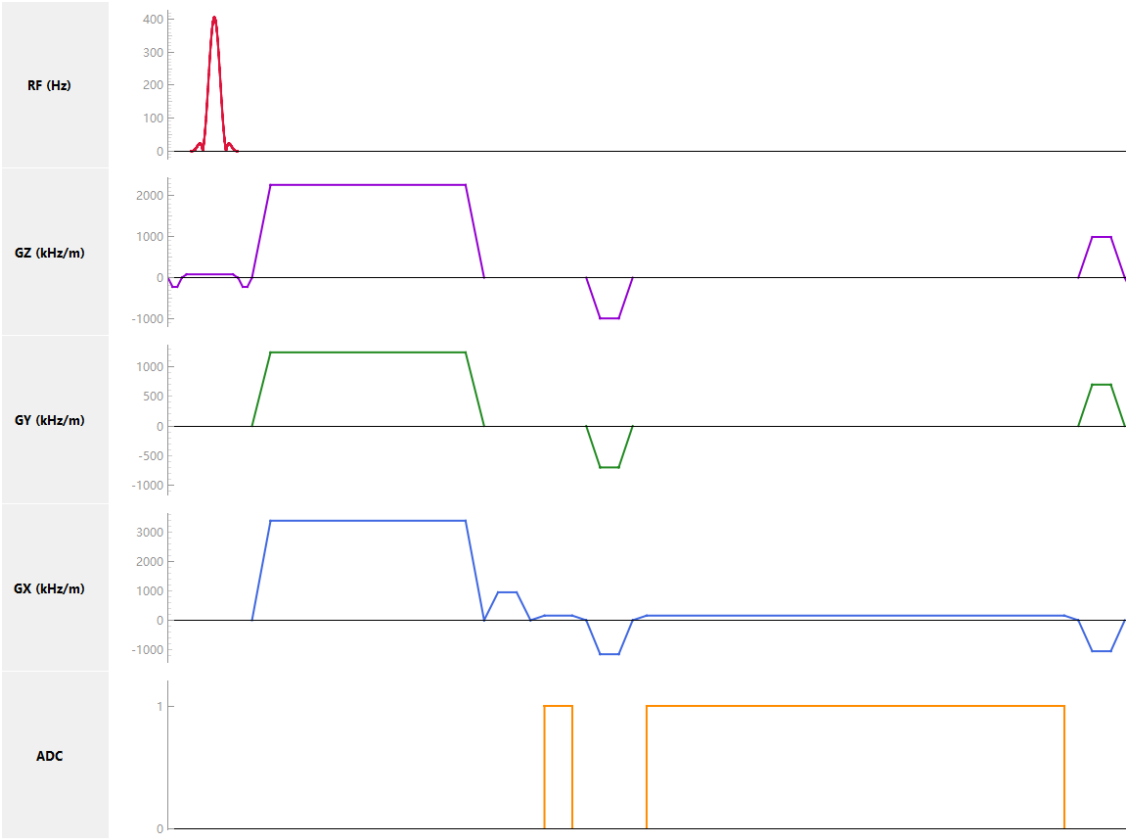
Top: TE_eff = 10ms, Bottom: TE_eff = 90ms

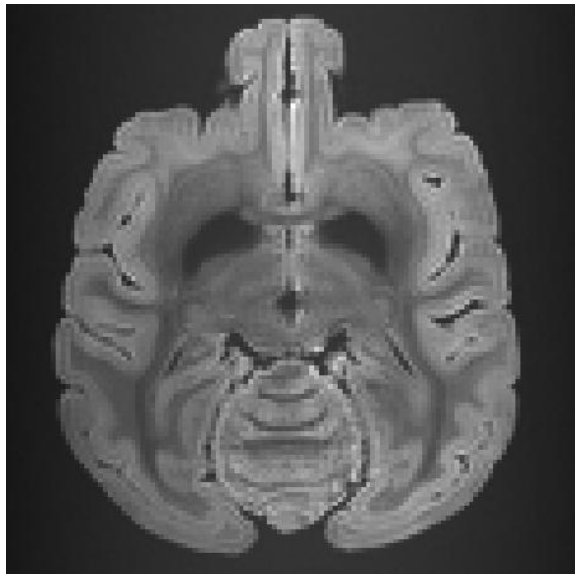
[1] Hennig et al., Magn Reson Med. 2022;10.1002/mrm.29224

Showcase 2

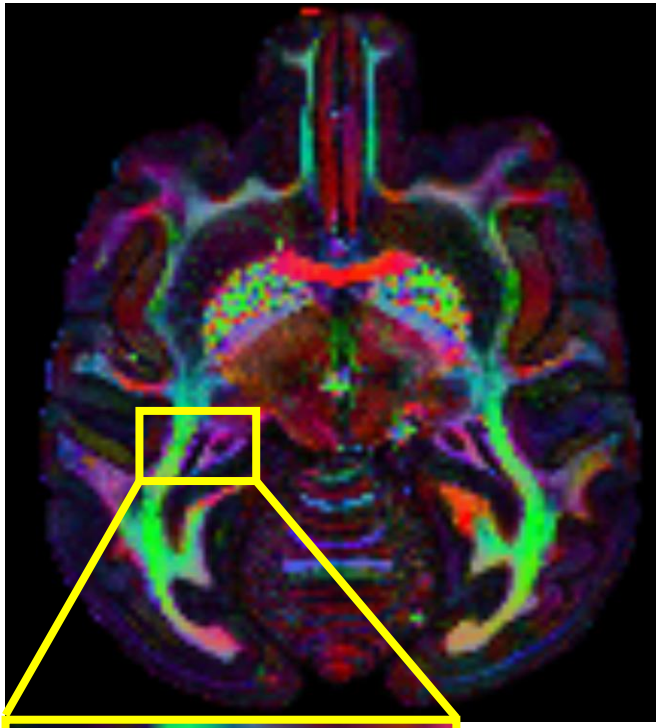
Self-navigated diffusion-weighted SSFP imaging of an ex-vivo macaque brain on the high-performance head-only 3T uMR Frontier system.

[flip angle=37°, Resolution=0.7×0.7×0.7mm, Matrix=150×100×70, TE=9.34ms, TR=20.61ms, BW=111Hz/pixel, b~5000s/mm^2, 32 directions]





Averaged diffusion weighted image (left) and colored fraction anisotropy (FA) map (right).

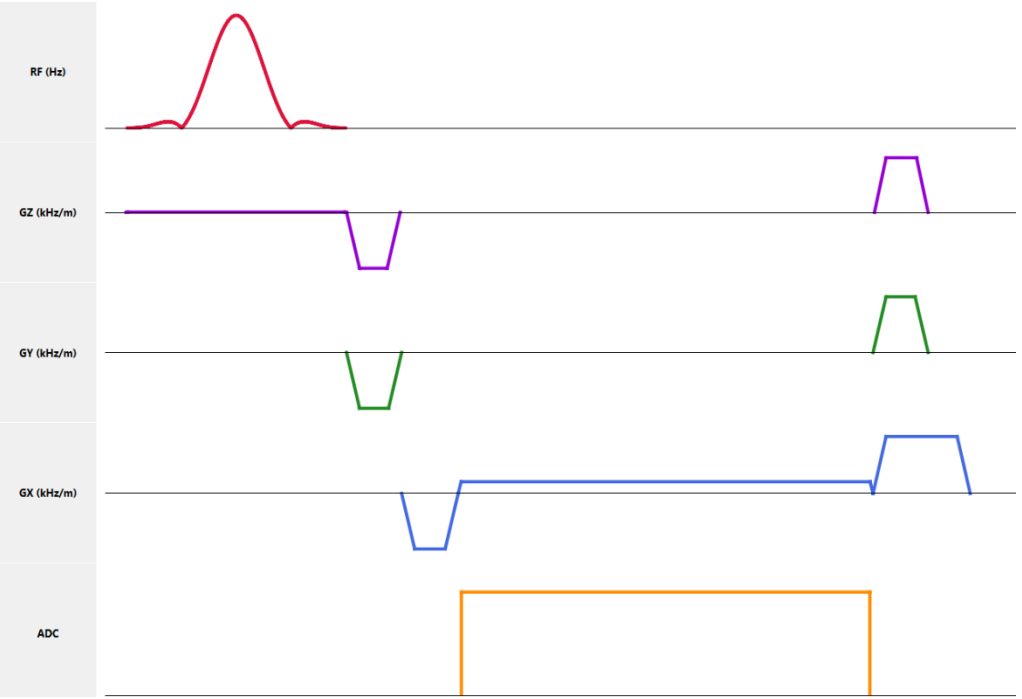


vof: occipital vertical fasciculus
ss: sagittal stratum
tap: tapetum of the corpus callosum
finfer: inferior forceps
cg: cingulum
strk: stratum calcarinum

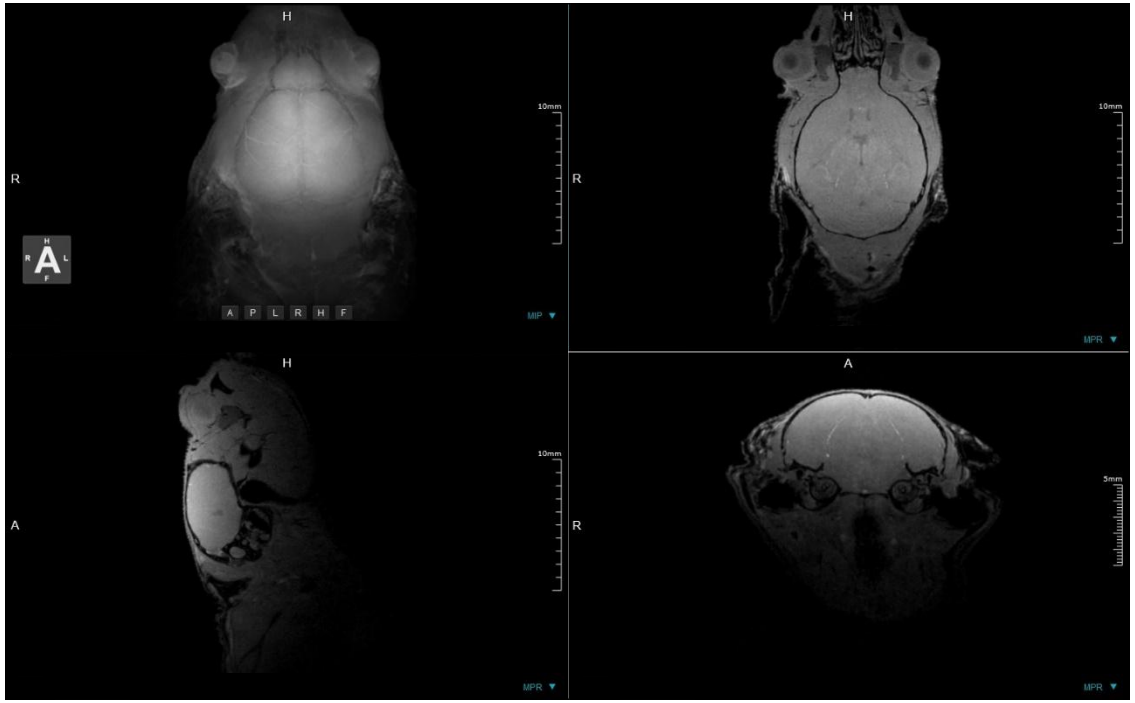
Showcase 3

Typical GRE 3D imaging in vivo at **uMR 9.4T** [mouse brain]

[flip angle=7°, Resolution=0.08×0.08×0.08mm, Matrix=256×256×380, TE=3ms, TR=10ms, BW=350Hz/pixel]



Sequence waveform in one TR

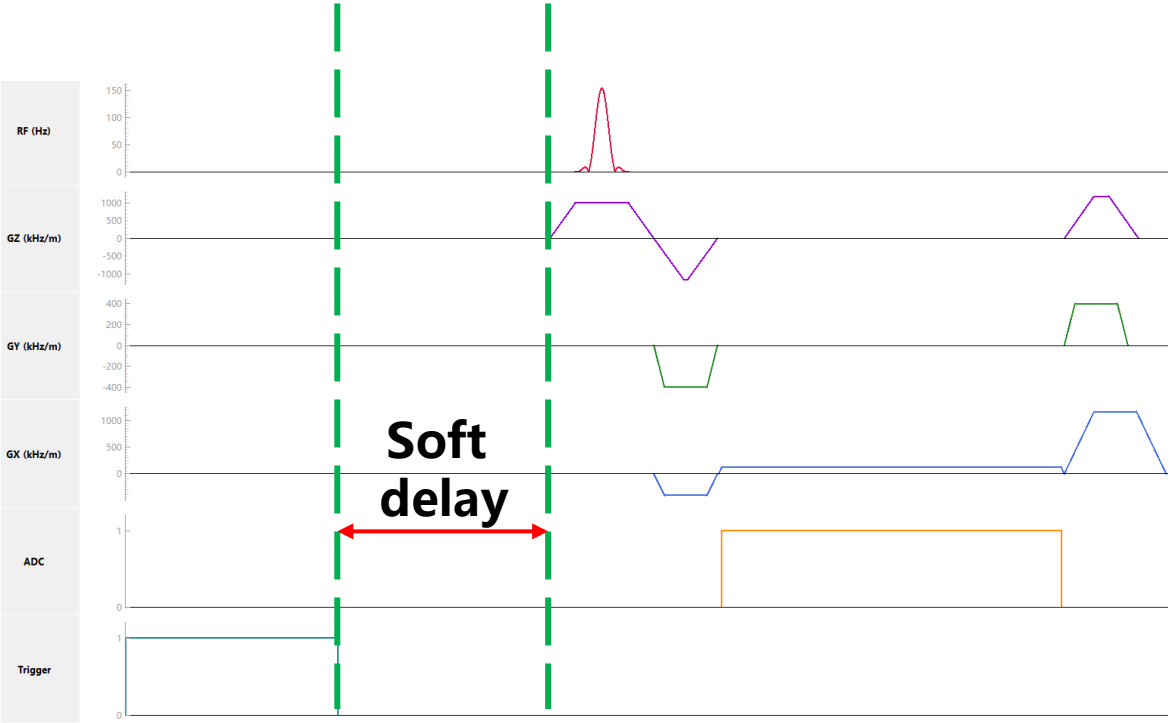


MPR images in coronal (upper right), sagittal (lower left), and transverse (lower right) orientations.

Showcase 4

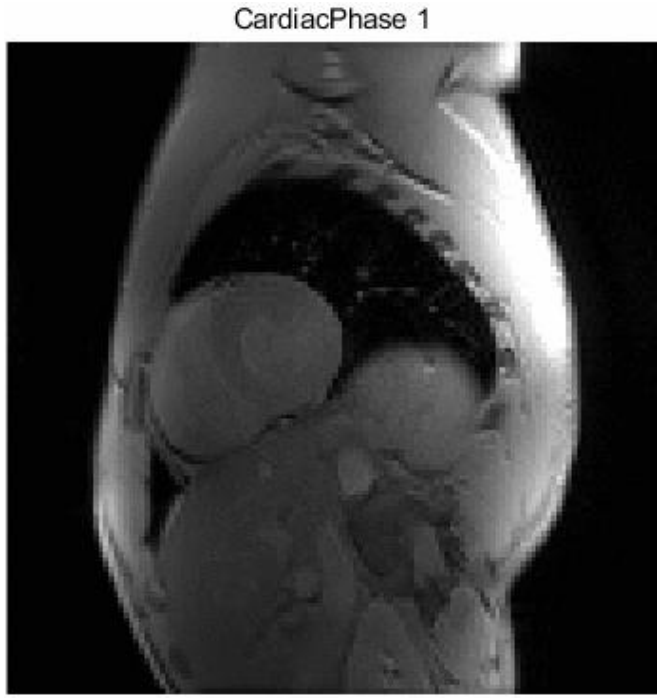
In-vivo cine pulseq sequence with **soft delay**, to enable scanning of subjects with **different RR-intervals**

[flip angle=7° FOV=320x320 mm, slice thickness=8mm TE=2.7ms, TR=5.8ms, BW=312Hz/pixel, Cardiac Phases=16]



Sequence waveform in one TR

Subject 1



Subject 2

Resources for UIH ADEPT

Install Sites

ADEPT Developer Forum*

*only available to NDA users for now

UIH ADEPT Workshop

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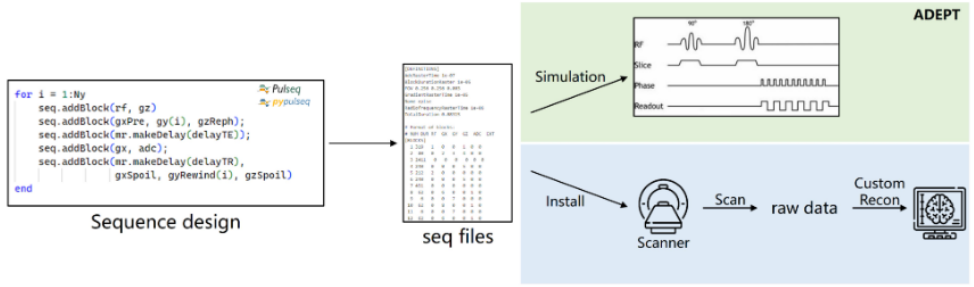
For latest information, please visit:

https://adept-forge.github.io

United Imaging MR ADEPT Platform

Pulseq on UIH MR

uMR ADEPT(Application Development Environment and Programming Tools) platform now fully supports the Pulseq open-source framework with the UIH Pulseq Interpreter. From the 3T MRI, to the pioneering 5T platform, to cutting-edge 9.4T ultra-high field systems, uMR ADEPT offers full compatibility, paving the way for your research innovations.



To obtain the Pulseq Interpreter on UIH MR, it is necessary for you to sign the **MR research collaboration agreement** with UIH MR.

If you have any interests in the Pulseq interpreter, please feel free to contact us: umr-adept@united-imaging.com.

Acknowledge

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PASSION *for* CHANGE